# GPIO library installation

**GPIO library installation**

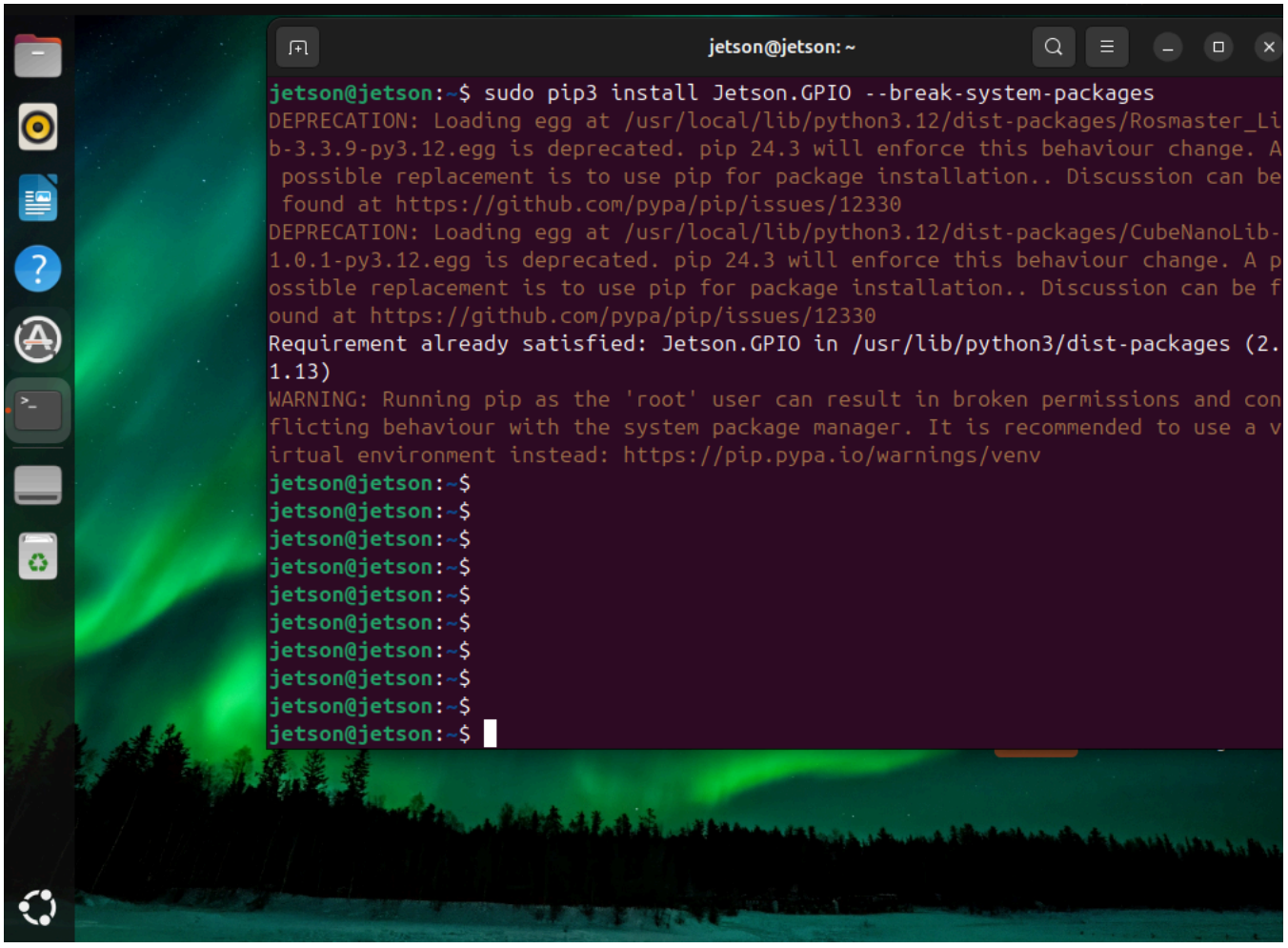
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## 1. Installing the Jetson.GPIO library

The Jetson.GPIO library is installed by default on the system, so this step can be skipped.

### 1.1. Automatic installation

```
sudo pip3 install Jetson.GPIO --break-system-packages
```



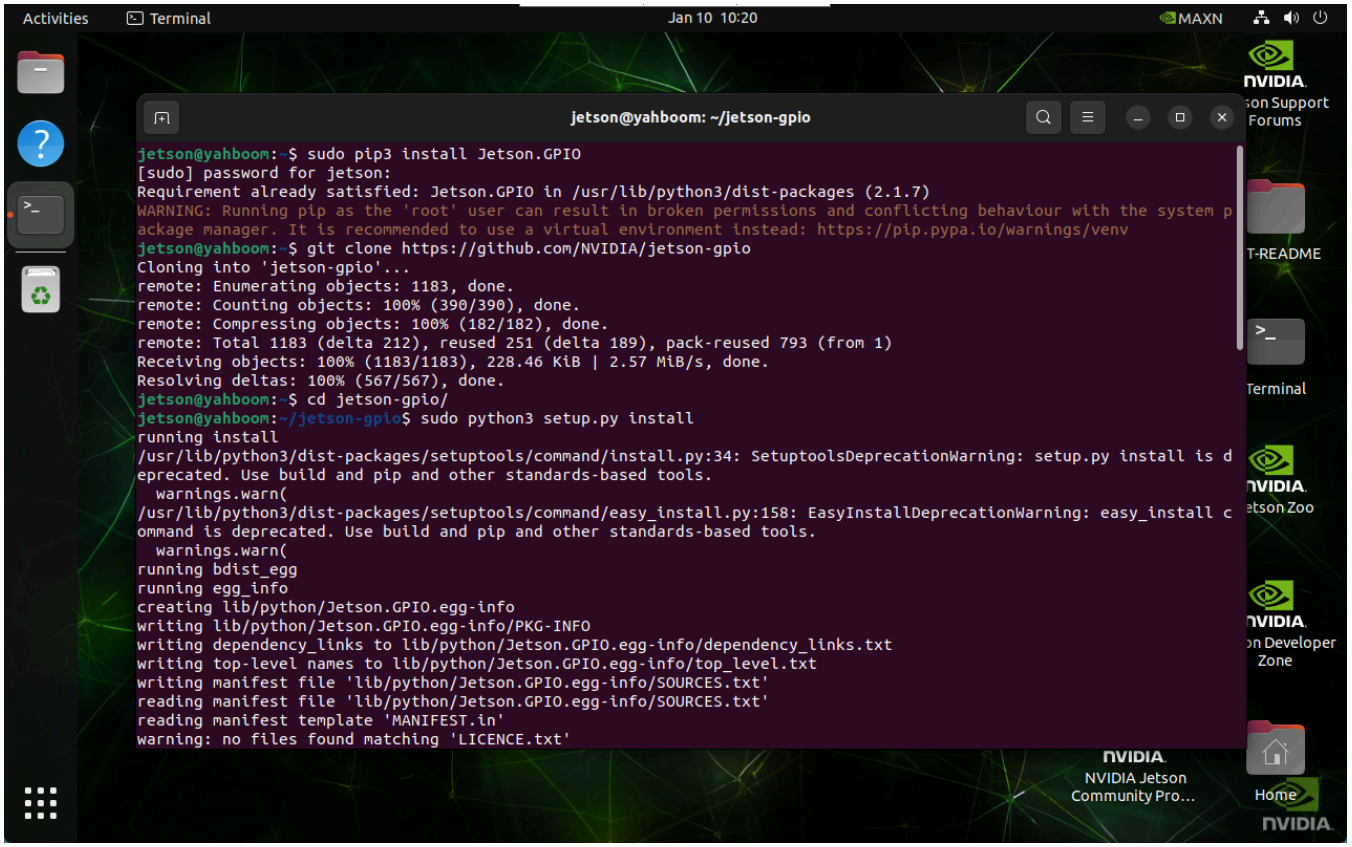
## 1.2. Manual installation

It is recommended to use the automatic installation method, because the manual installation may not be the latest version.

```
git clone https://github.com/NVIDIA/jetson-gpio
```

```
cd ~/jetson-gpio/
```

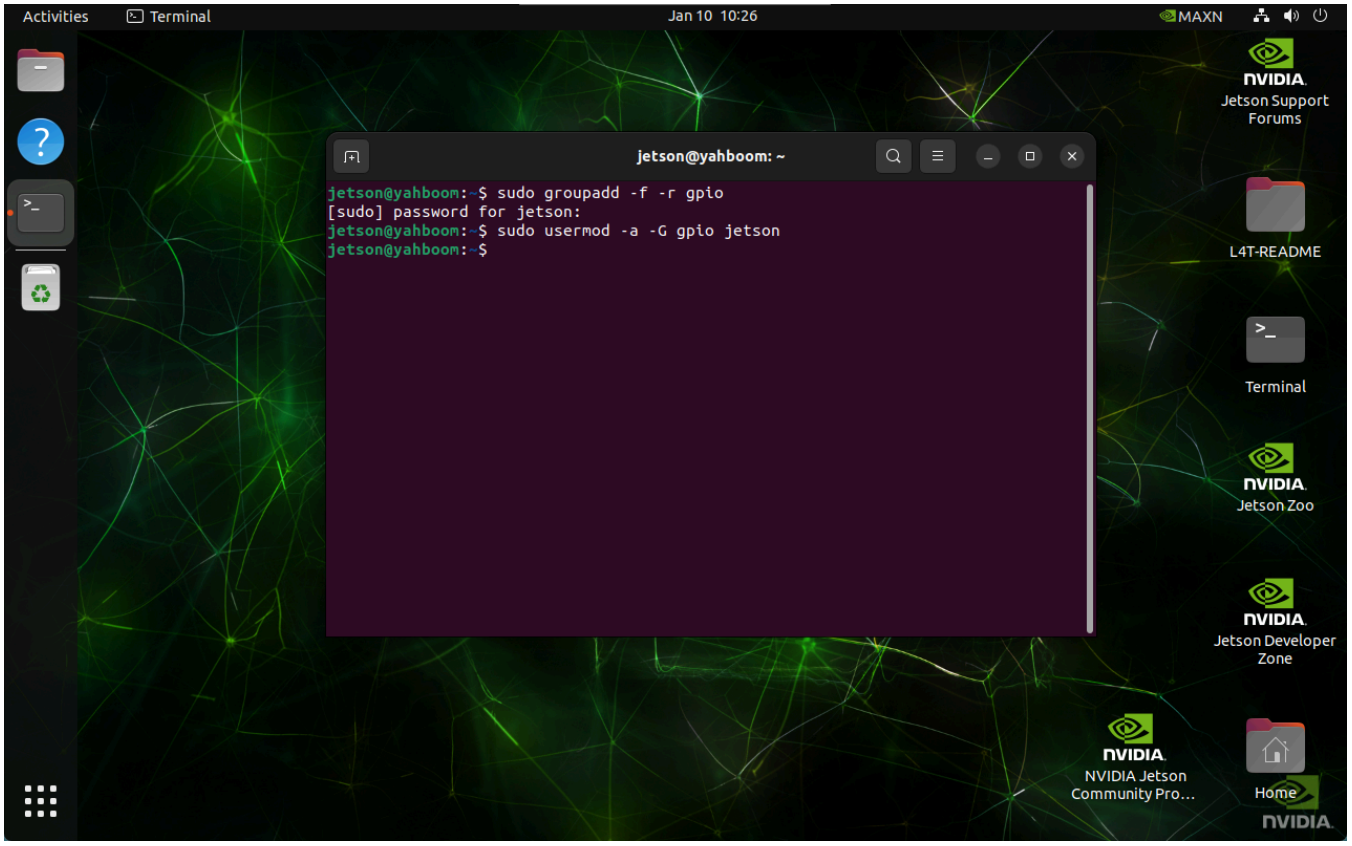
```
sudo python3 setup.py install
```




# 2. Setting the user permissions

Allow the current system user to access and use the Jetson.GPIO library: here `jetson` is the system user name.

```
sudo groupadd -f -r gpio
sudo usermod -a -G gpio jetson
```



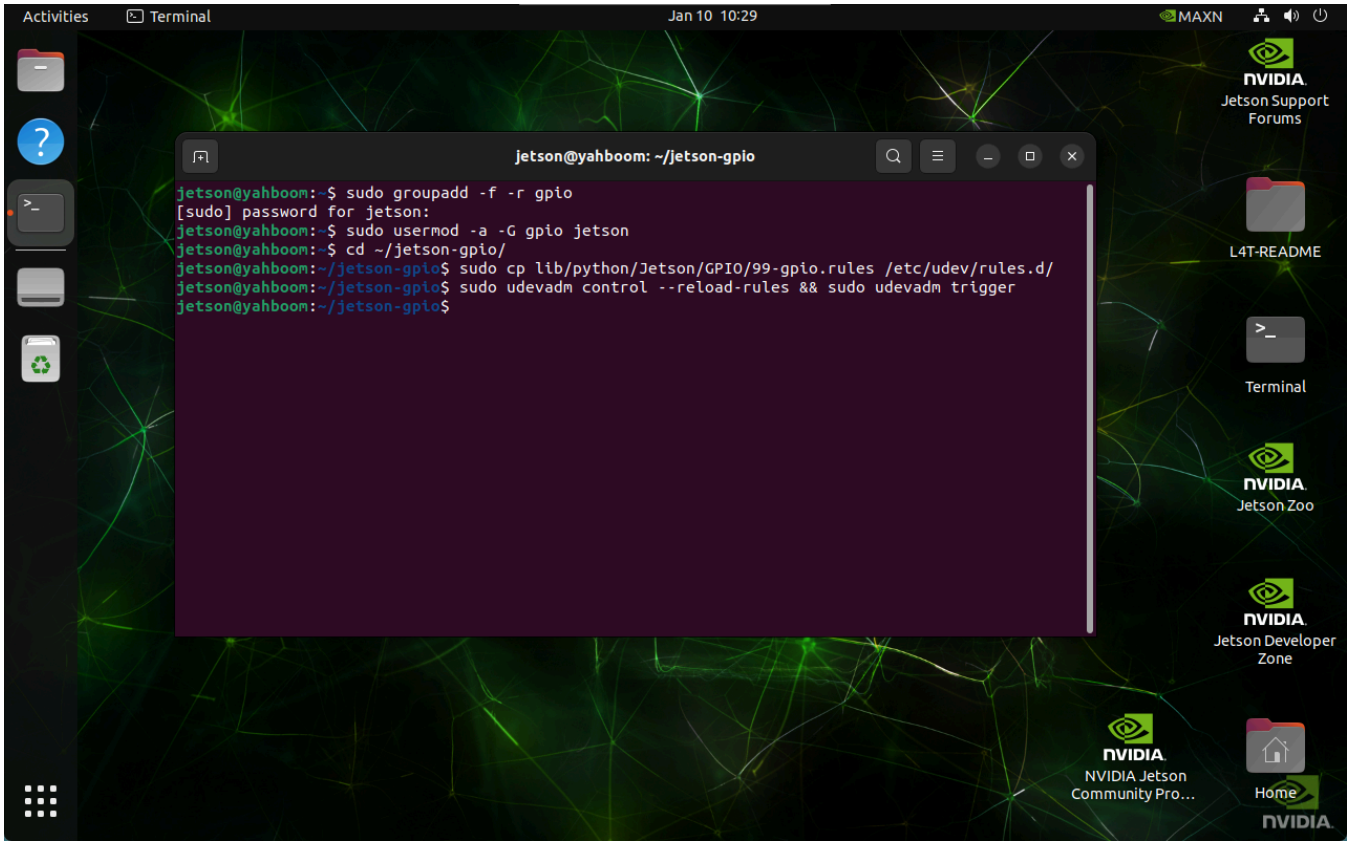

# 3. Custom rule file

## 3.1. Copy the rule file

```
cd ~/jetson-gpio/
sudo cp lib/python/Jetson/GPIO/99-gpio.rules /etc/udev/rules.d/
```

## 3.2. Reload the udev rules

```
sudo udevadm control --reload-rules && sudo udevadm trigger
```




# 4. Setting the mainboard model

Before controlling the GPIO, the mainboard model needs to be set in the terminal:

```
export JETSON_MODEL_NAME=JETSON_ORIN_NX
```

# 5. Reference material

https://github.com/NVIDIA/jetson-gpio